

Curves, common sheaves and Rao modules

Notes from Marcos Jardim's exercise sessions

IV Brazilian Workshop on Lefschetz Properties

UFBA, Salvador — September 28–October 1, 2026

These notes combine Larissa's session notes and the blackboard photographs. The common-plane argument was explained by Thiago Fassarella; the sheaf-Ext criteria were explained by Charles. The explanations between the calculations are added to make the arguments readable without the blackboard. Appendix A recalls the sheaf terminology and explains Ext. The monad in Section 4 remains a proposed construction with extra conditions to check.

The question running through the exercises is this: *when can two curves be obtained from sections of the same sheaf?* We work on projective three-space:

$$X = \mathbb{P}_{\mathbb{C}}^3, \quad R = \mathbb{C}[x_0, x_1, x_2, x_3], \quad h = c_1(\mathcal{O}_X(1)).$$

Write $\mathcal{O}(a) = \mathcal{O}_X(a)$, $F(a) = F \otimes \mathcal{O}(a)$ and

$$H_*^i(F) = \bigoplus_{t \in \mathbb{Z}} H^i(X, F(t)).$$

The star in $H_*^i(F)$ means that we keep the cohomology of *every twist*, with the integer t recording the degree. For a graded module N , our shift convention is $N[e]_t = N_{t+e}$. Sheaf duals are denoted by F^* and duals of vector spaces by $V^\vee$.

1 From a section to a curve

Let E be a rank-two vector bundle with $\det E = \mathcal{O}(c)$. The integer c records its first Chern class: $c_1(E) = ch$. A section of $E(a)$ is initially a map $\mathcal{O} \rightarrow E(a)$: it sends the constant section 1 to the chosen section s . Tensoring with $\mathcal{O}(-a)$ gives the equivalent map $s : \mathcal{O}(-a) \rightarrow E$. This is why the minus sign appears.

Suppose the zero scheme of s is a curve C , of pure codimension two. Then

$$0 \longrightarrow \mathcal{O}(-a) \xrightarrow{s} E \longrightarrow \mathcal{I}_C(c+a) \longrightarrow 0. \quad (1)$$

There are two things to understand here. First, where E is trivial, the section is a pair of functions $s = (f, g)$ and $C = V(f, g)$. The map

$$(u, v) \longmapsto fv - gu$$

has image the ideal (f, g) . Since f, g form a regular sequence, its kernel is generated by (f, g) . Thus quotienting by the section really does give the ideal of its zero curve. This also shows that C is locally a complete intersection.

Second, the twist of that ideal is forced by the determinant. If it were $\mathcal{I}_C(b)$, then away from C the determinant identity would be

$$\mathcal{O}(c) \cong \mathcal{O}(-a) \otimes \mathcal{O}(b).$$

Hence $b = c + a$. The ideal sheaf is torsion-free, so the inclusion of the section has *saturated* image. This does not say that its quotient is a line bundle along C .

The starting question is whether two curves can occur in sequences of the form (1) with the same middle sheaf. It matters whether this middle sheaf must be a vector bundle or may instead be reflexive. The distinction will be essential for the line and the conic.

What the Rao module measures. For a locally Cohen–Macaulay curve $C \subset X$, its *Rao module* is

$$M_C = H_*^1(\mathcal{I}_C).$$

It is a finite-length graded R -module. Multiplication by $f \in R_s$ sends its degree- t part to its degree- $(t + s)$ part through the map on cohomology induced by multiplication by f . The exact sequence

$$0 \longrightarrow \mathcal{I}_C(t) \longrightarrow \mathcal{O}(t) \longrightarrow \mathcal{O}_C(t) \longrightarrow 0$$

explains its meaning. For $t \geq 0$,

$$(M_C)_t \cong \text{coker}(R_t \longrightarrow H^0(C, \mathcal{O}_C(t))).$$

Thus it measures sections on C that do not extend to homogeneous polynomials on the ambient projective space. Its module structure remembers how these classes change under multiplication by polynomials, not just their dimensions. The curve is arithmetically Cohen–Macaulay exactly when $M_C = 0$.

Proposition 1.1. *If C_1, C_2 occur in (1) with a common E and twists a_1, a_2 , then*

$$M_{C_1}[c + a_1] \cong M_{C_2}[c + a_2], \quad M_{C_1} \cong M_{C_2}[a_2 - a_1].$$

Proof. For every integer m , $H^1(\mathcal{O}(m)) = H^2(\mathcal{O}(m)) = 0$. Twisting (1) by t therefore gives

$$0 \longrightarrow H^1(E(t)) \longrightarrow H^1(\mathcal{I}_{C_i}(t + c + a_i)) \longrightarrow 0.$$

The zero on the left is $H^1(\mathcal{O}(t - a_i))$; the zero on the right is $H^2(\mathcal{O}(t - a_i))$. Consequently

$$H^1(E(t)) \cong H^1(\mathcal{I}_{C_i}(t + c + a_i)).$$

These isomorphisms commute with multiplication by homogeneous polynomials. Both curves therefore give the same graded module $H_*^1(E)$, with different choices of where degree zero sits. $\square$

What Rao’s theorem says. Rao’s classification says that two locally Cohen–Macaulay curves in $\mathbb{P}^3$ belong to the same *even liaison class* if and only if their Rao modules are isomorphic up to shift [1]. Even liaison means a chain of an even number of complete-intersection links. This classification does not assert the existence of one common rank-two vector bundle in (1). A common bundle implies equality of shifted Rao modules; Rao converts that equality into a chain of links. The reverse passage to bundles allows changing rank by adding sums of line bundles, as well as twisting.

Higher-rank version. If the bundle has rank r , a single section generally leaves a quotient of rank $r - 1$. To obtain an ideal sheaf, which has rank one, we instead use $r - 1$ sections and sequences

$$0 \longrightarrow \mathcal{O}(-a_i)^{\oplus(r-1)} \longrightarrow E \longrightarrow \mathcal{I}_{C_i}(b_i) \longrightarrow 0, \quad b_i = c + (r - 1)a_i.$$

When the degeneracy locus has codimension two, it is defined locally by the maximal minors of the matrix of sections. The same proof gives

$$M_{C_1}[b_1] \cong M_{C_2}[b_2].$$

The classical stable correspondence allows sums of line bundles of different degrees and stabilization by such sums. It does not prescribe this particular rank and these particular kernels.

2 Common quotients and extensions

The next diagram is a way to compare two presentations without discarding the maps between the sheaves. Suppose E has two saturated line-sheaf inclusions $A_i = \mathcal{O}(-a_i) \rightarrow E$, with torsion-free quotients Q_i :

$$0 \longrightarrow A_i \xrightarrow{s_i} E \longrightarrow Q_i \longrightarrow 0. \quad (2)$$

Then

$$H_*^1(Q_1) \cong H_*^1(E) \cong H_*^1(Q_2).$$

There is no shift here unless the quotients themselves are retwisted. The converse asks whether equality of these cohomology modules can produce a common middle sheaf with the specified kernels.

Quotienting by both sections. We have $Q_1 = E/s_1(A_1)$ and $Q_2 = E/s_2(A_2)$. If we quotient by both images, in either order, we get

$$T = E/(s_1(A_1) + s_2(A_2)).$$

If the sections are independent over the function field, neither disappears when we quotient by the other. This gives the diagram

$$\begin{array}{ccccc} & & A_1 & \xlongequal{\quad} & A_1 \\ & & \downarrow s_1 & & \downarrow \\ A_2 & \xrightarrow{s_2} & E & \longrightarrow & Q_2 \\ \parallel & & \downarrow & & \downarrow \\ A_2 & \longrightarrow & Q_1 & \longrightarrow & T \end{array}$$

has four short exact rows and columns, with zeros at their ends understood. In particular, the bottom row and the right column say

$$Q_1/A_2 \cong T \cong Q_2/A_1.$$

The two inclusions into the quotients are the original sections followed by the quotient maps. A nonzero map from a line bundle to a torsion-free sheaf is injective. In particular, if $a_1 < a_2$, the map $A_1 \rightarrow Q_2$ cannot vanish: vanishing would factor s_1 through a map

$$A_1 \longrightarrow A_2, \quad \text{Hom}(A_1, A_2) = H^0(\mathcal{O}(a_1 - a_2)) = 0.$$

At $a_1 = a_2$, proportional sections must be excluded.

Reading the diagram backwards. Conversely, given surjections $Q_i \rightarrow T$ with kernels A_2, A_1 , respectively, the fibre product $E = Q_1 \times_T Q_2$ has both sequences (2). Its local sections are pairs (q_1, q_2) whose images in T agree. Projection onto Q_1 has kernel A_1 , and projection onto Q_2 has kernel A_2 . Whether this E is locally free or reflexive is a separate local question. The analogous graded-module construction uses injections $R[-a_i] \rightarrow M$ and their cokernels; sheafification gives the same diagram on projective space.

How this becomes ordinary linkage. In rank two, write $Q_i = \mathcal{I}_{C_i}(b_i)$. The determinant $s_1 \wedge s_2$ defines a surface S containing both curves. Here assume the sections are generically independent and, for simplicity, that S is smooth. The common quotient is

$$\mathcal{I}_{C_1/S}(b_1) \cong T \cong \mathcal{I}_{C_2/S}(b_2),$$

where we view sheaves on S as sheaves on projective space by extension by zero. On a smooth surface the ideal of a curve is the line bundle $\mathcal{O}_S(-C_i)$. Thus, if H_S is the hyperplane class of S ,

$$C_2 \sim C_1 + (b_2 - b_1)H_S.$$

This is an elementary *biliaison*. To see the two actual links, choose a general residual curve D such that $C_1 + D$ is a sufficiently high-degree hypersurface section of S . The displayed equivalence says that $C_2 + D$ is also a hypersurface section of S . Hence both unions are complete intersections in $\mathbb{P}^3$: first link C_1 to D , then D to C_2 . For example, if $S = V(F)$ and $C_1 \cup D = V(F, G)$, the first link is

$$I_D = (F, G) : I_{C_1}.$$

This is the residual-ideal construction. Hartshorne's Proposition 2.5 treats the sheaf argument using generalized divisors, so the surface need not be smooth [1].

Extension spaces. Sequences (2), with end terms fixed, are classified by $\text{Ext}^1(Q_i, A_i)$. An element of this group specifies how to assemble A_i and Q_i into a middle sheaf E . The zero element gives the direct sum; other elements give nonsplit extensions. Serre duality gives

$$\text{Ext}^1(Q_i, \mathcal{O}(-a_i)) \cong H^2(Q_i(a_i - 4))^\vee.$$

Thus an isomorphism of $H_*^1(Q_i)$ does not directly identify the extension spaces. Even an isomorphism of these spaces would not by itself identify the middle sheaves. Section 4 gives a concrete obstruction to the proposed converse.

3 A line and a conic

Let L be a line and Q a smooth plane conic in X . The line is cut out by two linear equations; the conic by one linear and one quadratic equation. Each pair has one relation, giving the resolutions

$$\begin{aligned} 0 &\longrightarrow \mathcal{O}(-2) \longrightarrow \mathcal{O}(-1)^{\oplus 2} \longrightarrow \mathcal{I}_L \longrightarrow 0, \\ 0 &\longrightarrow \mathcal{O}(-3) \longrightarrow \mathcal{O}(-1) \oplus \mathcal{O}(-2) \longrightarrow \mathcal{I}_Q \longrightarrow 0. \end{aligned}$$

For instance, if $I_L = (f, g)$, its relation is $-gf + fg = 0$. The coefficients have degree one, so the relation between the two degree-one generators has total degree two: this explains the $\mathcal{O}(-2)$ on the left. The conic has a relation of total degree three for the same reason. Taking cohomology and using the vanishing of H^1 and H^2 for line bundles gives $M_L = M_Q = 0$. Thus Rao's theorem puts them in the same even liaison class. We now ask the stronger question about a common sheaf.

Computing the extensions. Fixing a and c fixes the two end terms in (1). We can first ask how many extension classes there are. For $C = L$ or Q , Appendix A explains each step of

$$\begin{aligned} \text{Ext}^1(\mathcal{I}_C(c+a), \mathcal{O}(-a)) &\cong H^2(\mathcal{I}_C(c+2a-4))^\vee \\ &\cong H^1(\mathcal{O}_C(c+2a-4))^\vee. \end{aligned}$$

This reduces Ext on three-space to line-bundle cohomology on a curve. Both curves are isomorphic to $\mathbb{P}^1$, but their hyperplane bundles are different: a hyperplane meets L in one point and Q in

two. Thus $\mathcal{O}_L(m)$ corresponds to $\mathcal{O}_{\mathbb{P}^1}(m)$, whereas $\mathcal{O}_Q(m)$ corresponds to $\mathcal{O}_{\mathbb{P}^1}(2m)$. The usual formula on $\mathbb{P}^1$ gives

$$\begin{aligned} h^1(\mathcal{O}_L(m)) &= \max\{0, -m - 1\}, \\ h^1(\mathcal{O}_Q(m)) &= \max\{0, -2m - 1\}. \end{aligned}$$

Thus the two extension spaces have dimensions

$$\max\{0, 3 - c - 2a_1\}, \quad \max\{0, 7 - 2c - 4a_2\},$$

respectively. A nonzero class gives a nonsplit extension, not necessarily a locally free middle sheaf.

Proposition 3.1. *A line and a smooth conic cannot be zero schemes of sections of twists of one rank-two vector bundle.*

Proof. If such a bundle existed, adjunction would give

$$\omega_C \cong \mathcal{O}_C(c + 2a - 4).$$

The -4 comes from $\omega_{\mathbb{P}^3} = \mathcal{O}(-4)$, and $c + 2a$ from $\det(E(a))$. Both curves have canonical bundle $\mathcal{O}_{\mathbb{P}^1}(-2)$. For the line we must therefore have $c + 2a_1 - 4 = -2$. For the conic the twist is doubled on $\mathbb{P}^1$, so we must have $2(c + 2a_2 - 4) = -2$. Equivalently,

$$c + 2a_1 = 2, \quad c + 2a_2 = 3.$$

Subtracting gives $2(a_2 - a_1) = 1$, which is impossible. $\square$

The common plane (Thiago Fassarella's argument). We can still ask whether a rank-two reflexive sheaf might work: such a sheaf is allowed to fail to be a bundle at finitely many points. Suppose there is a rank-two *reflexive* sheaf E with $\det E = \mathcal{O}(c)$ and sequences

$$\begin{aligned} 0 \longrightarrow \mathcal{O}(-a_1) &\xrightarrow{s_1} E \longrightarrow \mathcal{I}_L(c + a_1) \longrightarrow 0, \\ 0 \longrightarrow \mathcal{O}(-a_2) &\xrightarrow{s_2} E \longrightarrow \mathcal{I}_Q(c + a_2) \longrightarrow 0. \end{aligned} \tag{3}$$

We will first find the degree of the determinant equation, then show that it vanishes on both curves.

Step 1: compare degrees through Chern classes. The total Chern class of a line bundle $\mathcal{O}(t)$ is $1 + th$. Total Chern classes multiply in exact sequences, so the two resolutions give

$$\begin{aligned} c(\mathcal{I}_L(t)) &= \frac{(1 + (t - 1)h)^2}{1 + (t - 2)h}, \\ c(\mathcal{I}_Q(t)) &= \frac{(1 + (t - 1)h)(1 + (t - 2)h)}{1 + (t - 3)h}. \end{aligned}$$

Expand these fractions and discard powers h^4 and higher, since they vanish on $\mathbb{P}^3$. The result is

$$\begin{aligned} c(\mathcal{I}_L(t)) &= 1 + th + h^2 + (2 - t)h^3, \\ c(\mathcal{I}_Q(t)) &= 1 + th + 2h^2 + (6 - 2t)h^3. \end{aligned}$$

Multiplying by $1 - a_i h$ in (3) and comparing the coefficients of h^2 gives

$$1 - a_1(c + a_1) = 2 - a_2(c + a_2),$$

or equivalently

$$\boxed{1 = (a_2 - a_1)(c + a_1 + a_2)}. \tag{4}$$

The 1 on the left is the difference between the degrees of the conic and the line. Both presentations must give the same $c_2(E)$; that is the only input in this equation.

Step 2: take the determinant of the two sections. The determinant of a reflexive sheaf is $(\wedge^2 E)^{**}$. Consequently the two sections give

$$s_1 \wedge s_2 \in H^0(\mathcal{O}(d)), \quad d = c + a_1 + a_2.$$

To track the twist, the wedge is a map from $\mathcal{O}(-a_1 - a_2)$ to $\det E = \mathcal{O}(c)$, so it is a homogeneous polynomial of degree d . Locally, where E is a bundle, it is just the determinant of the two columns s_1, s_2 . This section is nonzero. Otherwise the two saturated images would be the same line over the function field, and hence the same subsheaf of E . This would force $a_1 = a_2$, contrary to (4). Therefore $d \geq 0$, and (4) forces

$$d = 1, \quad a_2 - a_1 = 1.$$

The alternative factorization $1 = (-1)(-1)$ is ruled out because a negative-degree line bundle has no nonzero global sections. Thus the determinant is a nonzero linear polynomial, and its zero divisor is a plane:

$$\boxed{H = Z(s_1 \wedge s_2), \quad L \cup Q \subset H.}$$

To see the containment, compose s_2 with the quotient to $\mathcal{I}_L(c + a_1)$. This gives a section of $\mathcal{I}_L(d)$. The other composition gives a section of $\mathcal{I}_Q(d)$. Both are the determinant section, up to sign, so its linear equation belongs to both ideals. On the locally free locus this is just the determinant of two column vectors, which vanishes wherever either section vanishes.

Step 3: record the remaining constraints. Substituting $a_2 = a_1 + 1$ into $c + a_1 + a_2 = 1$ gives $c = -2a_1$. Using the line's sequence, we also get, writing $c_i(E)$ for its integer coefficient,

$$c = -2a_1, \quad c_2(E) = 1 + a_1^2, \quad c_3(E) = 2.$$

For the last equality, the coefficient of h^3 in the line's sequence is $2 - (c + a_1) - a_1 = 2 - c - 2a_1 = 2$. The third Chern class again rules out local freeness. These are necessary conditions for (3); they do not yet construct the common sheaf.

The trial used in the exercises. For $a_1 = 1, a_2 = 2, c = -2$,

$$\begin{aligned} 0 &\longrightarrow \mathcal{O}(-1) \longrightarrow E \longrightarrow \mathcal{I}_L(-1) \longrightarrow 0, \\ 0 &\longrightarrow \mathcal{O}(-2) \longrightarrow E \longrightarrow \mathcal{I}_Q \longrightarrow 0. \end{aligned}$$

Here $c_2(E) = 2$ and both extension spaces have dimension three: they reduce to $H^1(\mathcal{O}_L(-4))$ and $H^1(\mathcal{O}_Q(-2))$, which both identify, via the parametrizations, with $H^1(\mathcal{O}_{\mathbb{P}^1}(-4))$. The induced maps are $\mathcal{O}(-1) \rightarrow \mathcal{I}_Q$ and $\mathcal{O}(-2) \rightarrow \mathcal{I}_L(-1)$. They are multiplication by the equation of the common plane H . Quotienting by this equation means working inside H . There, a line has ideal $\mathcal{O}_H(-1)$ and a conic has ideal $\mathcal{O}_H(-2)$. The extra twist on the line makes the two quotients agree. For $i : H \hookrightarrow X$, this says

$$T \cong i_* \mathcal{I}_{L/H}(-1) \cong i_* \mathcal{O}_H(-2) \cong i_* \mathcal{I}_{Q/H}.$$

This is the rank-two instance of the quotient diagram in Section 2.

More generally, $H^0(\mathcal{I}_C(k)) \neq 0$, for $k > 0$, means that C lies in a degree- k hypersurface: a nonzero section is a homogeneous polynomial vanishing on C . For a coplanar line and conic with equations $L = V(\ell, \ell')$ and $Q = V(\ell, q)$, their union is $V(\ell, \ell'q)$ and has degree three.

Dualizing and locating singularities. For $C = L, Q$ and $t = c + a$, dualizing the corresponding sequence gives

$$\begin{aligned} 0 &\longrightarrow \mathcal{O}(-t) \longrightarrow E^* \longrightarrow \mathcal{O}(a) \\ &\longrightarrow \mathcal{E}xt^1(\mathcal{I}_C(t), \mathcal{O}) \longrightarrow \mathcal{E}xt^1(E, \mathcal{O}) \longrightarrow 0. \end{aligned}$$

The curve resolutions identify

$$\begin{aligned} \mathcal{E}xt^1(\mathcal{I}_L(t), \mathcal{O}) &\cong \mathcal{O}_L(2 - t), \\ \mathcal{E}xt^1(\mathcal{I}_Q(t), \mathcal{O}) &\cong \mathcal{O}_Q(3 - t). \end{aligned}$$

Hence the support of $\mathcal{E}xt^1(E, \mathcal{O})$ is contained in $L \cap Q$: this sheaf is a quotient of a sheaf supported on L , and also a quotient of a sheaf supported on Q . For reflexive E , its support is exactly the set where E is not locally free. So any singular points of the common sheaf must lie in the intersection of the two curves. Appendix A explains why sheaf Ext detects this failure.

4 Higher rank: an obstruction and a monad

Equal cohomology is insufficient. The problem with trying to reconstruct a sheaf from H_*^1 is that line-bundle summands are invisible to this module. They have no H^1 in any twist, but they do change Chern classes. A concrete version of the session's Chern-class example uses the cotangent bundle $F = \Omega_X^1$ and

$$Q_1 = F \oplus \mathcal{O}^{\oplus 2}, \quad Q_2 = F \oplus \mathcal{O}(1) \oplus \mathcal{O}(-1).$$

The Euler sequence is

$$0 \longrightarrow \Omega_X^1 \longrightarrow \mathcal{O}(-1)^{\oplus 4} \longrightarrow \mathcal{O} \longrightarrow 0.$$

Thus $c(F) = (1 - h)^4$ modulo h^4 , and

$$\begin{aligned} c(Q_1) &= 1 - 4h + 6h^2 - 4h^3, \\ c(Q_2) &= c(F)(1 + h)(1 - h) = 1 - 4h + 5h^2. \end{aligned}$$

Yet

$$H_*^1(Q_1) \cong H_*^1(F) \cong H_*^1(Q_2),$$

since line bundles have no H_*^1 . If both appeared in (2) with a common E , then

$$(1 - a_1 h)c(Q_1) = (1 - a_2 h)c(Q_2).$$

The coefficient of h forces $a_1 = a_2$, and the coefficient of h^2 then gives a contradiction. This disproves the converse for these fixed quotients and line-bundle kernels.

Starting from a module. We can also run the construction in the other direction: start with a module and seek a sheaf having that module as H_*^1 . Let M be a nonzero finite-length graded R -module, with a minimal free presentation

$$L_1 \xrightarrow{\beta} L_0 \longrightarrow M \longrightarrow 0, \quad L_0 = \bigoplus_i R[-a_i]^{\oplus m_i}.$$

The tilde takes a graded module to its associated sheaf on $\text{Proj } R$. A finite-length graded module is supported at the irrelevant ideal (x_0, x_1, x_2, x_3) , which is not a point of projective space. Consequently $\tilde{M} = 0$, so $\tilde{\beta} : \tilde{L}_1 \rightarrow \tilde{L}_0$ is surjective. Its kernel K is a vector bundle: locally a surjection onto a free module splits. We have

$$0 \longrightarrow K \longrightarrow \tilde{L}_1 \longrightarrow \tilde{L}_0 \longrightarrow 0.$$

The long exact cohomology sequence, in all twists, now gives

$$L_1 \longrightarrow L_0 \longrightarrow H_*^1(K) \longrightarrow 0.$$

Thus $H_*^1(K) \cong M$. This first step already produces a bundle with the desired cohomology; its rank and Chern classes may not be the ones we want.

What the monad would add. The further construction discussed in the session asks for an injection

$$\alpha : \tilde{L}_0^* \longrightarrow \tilde{L}_1, \quad \tilde{\beta}\alpha = 0.$$

The equation $\tilde{\beta}\alpha = 0$ means that the image of α lies inside K . The resulting complex

$$\tilde{L}_0^* \xrightarrow{\alpha} \tilde{L}_1 \xrightarrow{\tilde{\beta}} \tilde{L}_0$$

is called a *monad*: it is exact at the two ends, and its middle cohomology is

$$E = \ker \tilde{\beta} / \operatorname{im} \alpha.$$

Equivalently,

$$0 \longrightarrow \tilde{L}_0^* \longrightarrow K \longrightarrow E \longrightarrow 0.$$

Because the left term is a sum of line bundles, it has no H_*^1 or H_*^2 . Thus

$$H_*^1(E) \cong H_*^1(K) \cong \operatorname{coker}(L_1 \rightarrow L_0) \cong M.$$

The needed map α is additional data: dualizing β lands in $\tilde{L}_1^*$, not automatically in $\tilde{L}_1$. For E to be locally free, the image of α must be a subbundle of K . The ranks and first Chern classes are

$$\begin{aligned} \operatorname{rk} E &= \operatorname{rk} L_1 - 2 \operatorname{rk} L_0, \\ c_1(E) &= c_1(\tilde{L}_1). \end{aligned}$$

In the second equality the first Chern classes of $\tilde{L}_0$ and its dual cancel. Hence the intended condition $c_1(E) = 0$ must be checked on $\tilde{L}_1$. Constructing this monad and suitable saturated sections would then produce curves whose Rao module is M , up to shift.

5 The Lefschetz question

For a finite-length graded R -module N , the *weak Lefschetz property* asks for $\ell \in R_1$ such that

$$\times \ell : N_t \longrightarrow N_{t+1}$$

has maximal rank for every t . The *strong Lefschetz property* asks that the same ℓ make

$$\times \ell^s : N_t \longrightarrow N_{t+s}$$

have maximal rank for every t and $s \geq 1$. Maximal rank means that the rank is the smaller of the dimensions of the source and target. Thus the map must be injective when the source is smaller, and surjective when the target is smaller. One choice of ℓ must work in all degrees; for the strong property it must also work for every power. Jardim's question is which curves have Rao modules with these properties. Both properties survive graded isomorphisms and shifts, hence are invariants of the even liaison class. For ACM curves the Rao module is zero, so both hold vacuously.

A Sheaf primer

Here sheaves are coherent. Statements about rank and torsion use an integral variety; the Ext tests at the end use a smooth projective threefold.

Coherent, locally free, bundle. A coherent sheaf is locally described by finitely many generators and relations. On an affine open set $\text{Spec } R$, it corresponds to a finite R -module. It is *locally free of rank r* if every point has a neighbourhood U with

$$F|_U \cong \mathcal{O}_U^{\oplus r}.$$

This means that its sections locally have a basis of r elements. It is exactly the sheaf of sections of an algebraic vector bundle; a basis of sections is a frame of the bundle. Rank-one bundles are line bundles, also called invertible sheaves. The rank of an arbitrary coherent sheaf is its dimension over the function field at the generic point. It describes the sheaf on a dense open set, and does not guarantee a basis near every point.

Torsion concerns multiplication by scalars. Over a domain R , an element $m \in M$ is torsion if $fm = 0$ for some nonzero $f \in R$. The module is torsion-free if this forces $m = 0$. For example, take

$$R = \mathbb{C}[x], \quad M = R/(x).$$

The element $\bar{1} \in M$ is nonzero, but $x\bar{1} = 0$. The scalar x is nonzero in R ; its class in the quotient is zero. That is precisely how a nonzero scalar kills a nonzero module element. There are no nonzero nilpotents in $M \cong \mathbb{C}$. Torsion concerns the R -action on M , whereas nilpotence concerns powers of an element inside a ring.

For sheaves, apply the module definition to each stalk. If Z is a proper closed subscheme of an integral variety, then

$$0 \longrightarrow \mathcal{I}_Z \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Z \longrightarrow 0$$

has a torsion-free ideal sheaf: it is a submodule of a domain at each stalk. Its quotient is torsion as a sheaf on X , since local equations of Z annihilate it.

The canonical map to the double dual. The dual is

$$F^* = \mathcal{H}om(F, \mathcal{O}_X).$$

For modules, an element of M^* is an R -linear map $\varphi : M \rightarrow R$. Given $m \in M$, we can evaluate every such map at m ; this gives

$$\begin{aligned} \text{ev} : M &\longrightarrow M^{**}, \\ \text{ev}(m)(\varphi) &= \varphi(m). \end{aligned}$$

This is canonical because it makes no choice of basis. For a finite-dimensional vector space it is an isomorphism. For a module over a ring it may fail to be injective or surjective. A sheaf is *reflexive* precisely when this evaluation map is an isomorphism. For coherent sheaves on an integral Noetherian scheme, torsion-freeness is equivalent to its being injective [3].

The implication from reflexive to torsion-free has a short proof. Suppose $fm = 0$ with $f \neq 0$. For every $\varphi \in M^*$,

$$f\varphi(m) = \varphi(fm) = 0.$$

The ring is a domain, so $\varphi(m) = 0$ for every φ . Thus $\text{ev}(m) = 0$, and injectivity of evaluation gives $m = 0$. In particular, locally free implies reflexive, and reflexive implies torsion-free.

The reverse implications fail in three dimensions. For a line $L \subset \mathbb{P}^3$,

$$\mathcal{I}_L^{**} = \mathcal{O} \neq \mathcal{I}_L.$$

Thus its ideal sheaf is torsion-free but not reflexive. A reflexive sheaf on a smooth surface is locally free; on a smooth threefold it can fail to be locally free at finitely many points.

Twists: the arithmetic of degrees. Twisting means tensoring with a line bundle:

$$F(a) = F \otimes \mathcal{O}(a), \quad F(a)(b) = F(a + b).$$

It preserves exact sequences and rank. Two useful sign checks are

$$\begin{aligned} \operatorname{Hom}(\mathcal{O}(-a), F) &= H^0(F(a)), \\ \operatorname{Hom}(\mathcal{O}(a), \mathcal{O}(b)) &= H^0(\mathcal{O}(b - a)). \end{aligned}$$

The second map is multiplication by a homogeneous polynomial of degree $b - a$. It must be zero if $b - a < 0$. This is the same degree bookkeeping as the shifts $R[-a]$ in a graded free resolution: its generator has degree a . For a rank- r bundle, the determinant picks up the twist ra , because each of the r factors in a top wedge picks up one copy of $\mathcal{O}(a)$.

An inclusion is *saturated* if its quotient is torsion-free. Multiplication by a linear equation ℓ gives an injection $\mathcal{O}(-1) \rightarrow \mathcal{O}$, but its quotient is $\mathcal{O}_H$, where $H = V(\ell)$. This quotient has torsion on projective space, so the inclusion is not saturated. In contrast, the quotient in (1) is an ideal sheaf, hence torsion-free.

Chern classes in the calculations. These are the same characteristic classes used in differential geometry. For a complex vector bundle, $c_i(E) \in H^{2i}(X, \mathbb{Z})$. On $\mathbb{P}^3$, all we need is

$$H^*(\mathbb{P}^3, \mathbb{Z}) = \mathbb{Z}[h]/(h^4).$$

Write $c_1(E) = ch$ and $c_2(E) = eh^2$. For rank two,

$$\begin{aligned} c_1(E(a)) &= (c + 2a)h, \\ c_2(E(a)) &= (e + ac + a^2)h^2. \end{aligned}$$

To remember the second formula, use formal Chern roots: twisting adds ah to both roots. Expanding their product gives the terms e , ac and a^2 . For a regular section of $E(a)$ with curve zero scheme C , its zero-locus class is the top Chern class:

$$[C] = c_2(E(a)), \quad \deg C = e + ac + a^2.$$

For coherent sheaves on smooth projective space, Chern classes are defined using finite resolutions by bundles. Their total Chern classes multiply in short exact sequences. This is why we divide the Chern polynomials in the line and conic calculations. A rank-two bundle has $c_3 = 0$. A rank-two reflexive sheaf can have $c_3 \neq 0$: its class comes from an alternating sum of bundles in a resolution, not from one bundle of rank two. Here nonzero c_3 is an obstruction to local freeness.

Global Ext: assembling a middle sheaf and detecting splitting. The group $\operatorname{Ext}^1(B, A)$ classifies short exact sequences

$$0 \longrightarrow A \longrightarrow E \longrightarrow B \longrightarrow 0,$$

up to isomorphisms that are the identity on A and B . The zero class is the split sequence, with $E = A \oplus B$. Applying $\operatorname{Hom}(-, A)$ gives part of a long exact sequence:

$$\operatorname{Hom}(E, A) \longrightarrow \operatorname{Hom}(A, A) \xrightarrow{\delta} \operatorname{Ext}^1(B, A).$$

The image $\delta(1_A)$ is the extension class. Exactness says that it vanishes exactly when 1_A extends to a map $E \rightarrow A$. That map is a retraction of $A \rightarrow E$, so the sequence splits. This explains how Ext enters when we apply Hom to a short exact sequence.

Over a ring, one can compute Ext by resolving the first argument by projective modules, or the second by injective modules. These give the same groups. For sheaves, a locally free resolution computes a sheaf Hom complex; global Ext also requires its hypercohomology. In particular, if E is locally free, then

$$\operatorname{Ext}^i(E, A) \cong H^i(X, E^* \otimes A).$$

Local freeness of the first argument therefore does not force global Ext to vanish.

The Ext-to-cohomology calculation. On $\mathbb{P}^3$, Serre duality says

$$\mathrm{Ext}^1(B, A) \cong \mathrm{Ext}^2(A, B(-4))^\vee.$$

The degree 1 becomes $3 - 1 = 2$; the arguments exchange places; and $\omega_X = \mathcal{O}(-4)$ contributes the twist -4 . Insert $B = \mathcal{I}_C(c + a)$ and $A = \mathcal{O}(-a)$:

$$\begin{aligned} \mathrm{Ext}^1(\mathcal{I}_C(c + a), \mathcal{O}(-a)) &\cong \mathrm{Ext}^2(\mathcal{O}(-a), \mathcal{I}_C(c + a - 4))^\vee \\ &\cong H^2(\mathcal{I}_C(c + 2a - 4))^\vee. \end{aligned}$$

The last step is useful because the first argument is now a line bundle. Its dual is $\mathcal{O}(a)$; tensoring by that dual supplies the second a . Thus all the twists are accounted for.

Next use $0 \rightarrow \mathcal{I}_C(m) \rightarrow \mathcal{O}(m) \rightarrow \mathcal{O}_C(m) \rightarrow 0$. The long exact cohomology sequence contains

$$0 \longrightarrow H^1(\mathcal{O}_C(m)) \longrightarrow H^2(\mathcal{I}_C(m)) \longrightarrow 0,$$

since $H^1(\mathcal{O}(m))$ and $H^2(\mathcal{O}(m))$ vanish on $\mathbb{P}^3$ for every m . So the two middle groups are isomorphic. For a line or conic this leaves a calculation on $\mathbb{P}^1$, where

$$\begin{aligned} h^1(\mathcal{O}_{\mathbb{P}^1}(n)) &= h^0(\mathcal{O}_{\mathbb{P}^1}(-n - 2)) \\ &= \max\{0, -n - 1\}. \end{aligned}$$

The vanishing used above concerns *intermediate cohomology of line bundles on projective three-space*. It says nothing of that kind for arbitrary sheaves or for top cohomology. Indeed $H^3(\mathcal{O}(-4)) \cong \mathbb{C}$.

Sheaf Ext: a local test. The sheaf $\mathcal{E}xt^i(F, G)$ has stalks

$$\mathcal{E}xt^i(F, G)_x \cong \mathrm{Ext}_{\mathcal{O}_{X,x}}^i(F_x, G_x).$$

One computes it by applying sheaf Hom to a locally free resolution and taking the cohomology sheaves of that complex. Equivalently, it is the sheafification of the presheaf of Ext groups on open sets. It records local information, including where its stalks are nonzero.

If F is a bundle, its stalks are free modules, so its positive sheaf Ext vanishes. This is different from global Ext: for example,

$$\mathrm{Ext}^3(\mathcal{O}, \mathcal{O}(-4)) \cong H^3(\mathcal{O}(-4)) \cong \mathbb{C},$$

although the corresponding positive sheaf Ext groups vanish.

Charles's local-freeness and reflexivity criteria. Let F be torsion-free on a smooth projective threefold, and abbreviate

$$\mathcal{E}^i = \mathcal{E}xt^i(F, \mathcal{O}_X).$$

Then the precise statements are:

- F is locally free if and only if $\mathcal{E}^1 = \mathcal{E}^2 = 0$.
- F is reflexive if and only if $\mathcal{E}^2 = 0$ and $\mathcal{E}^1$ has finite support.

Finite support permits a nonzero stalk at finitely many points, or none. Thus, once we know that F is reflexive, $\mathcal{E}^1 = 0$ is exactly the test for local freeness. A reflexive sheaf may fail this test at isolated points, but not along a curve.

For the sheaves in the line-and-conic exercise, the ideal resolutions and the extension by a line bundle give local free resolutions of length at most one. Hence $\mathcal{E}^2 = 0$ is automatic there; finite support of $\mathcal{E}^1$ tests reflexivity. For a general torsion-free sheaf the condition on $\mathcal{E}^2$ must also be checked.

The general support criterion for reflexivity is [2]

$$\operatorname{codim}_X \operatorname{Supp} \mathcal{E}xt^i(F, \mathcal{O}_X) \geq i + 2 \quad (i > 0),$$

with F torsion-free. For $i = 1$ the support must have codimension at least three, so on a projective threefold it is finite. For $i = 2$ it would need codimension at least four, so it must be empty. Locally, Auslander–Buchsbaum says

$$\operatorname{pd} F_x + \operatorname{depth} F_x = \dim \mathcal{O}_{X,x}.$$

Projective dimension is the length of a shortest free resolution of the stalk. Torsion-freeness gives positive depth at points of positive codimension, so this length is at most two here. The highest nonzero local Ext detects this length. This explains why only $\mathcal{E}^1$ and $\mathcal{E}^2$ are needed in the two tests.

Finally, the ideal sheaf of a point shows why checking only first Ext can give the wrong answer:

$$\mathcal{E}xt^1(\mathcal{I}_p, \mathcal{O}) = 0, \quad \mathcal{E}xt^2(\mathcal{I}_p, \mathcal{O}) \cong \mathcal{O}_p.$$

Nevertheless $\mathcal{I}_p$ is neither locally free nor reflexive.

References

- [1] R. Hartshorne, *On Rao's Theorems and the Lazarsfeld–Rao Property*, 2003. arXiv:math/0302078.
- [2] D. Huybrechts and M. Lehn, *The Geometry of Moduli Spaces of Sheaves*, Section 1.1, especially Propositions 1.1.6 and 1.1.10.
- [3] The Stacks Project, Tag 0AV0 and Tag 0AVT, on reflexive modules and sheaves.